Linear Generalization vs. Random Memorization at Exactly Matched Frequency

A Behavioral and Mechanistic Audit

factual-recall-mechanisms

2026-07-13

1 Question and corrected result in one paragraph

One small transformer is trained for **exactly one epoch** on a synthetic corpus containing two relations with *identical frequency statistics at every level* (same subjects, same 32 candidate objects, same number of triples, same number of occurrences per triple). The relations differ only in their semantics: R_{sem} maps each subject to an object determined by the subject's f-attribute, whereas R_{rnd} maps it to a balanced random object. At 64 occurrences per triple, both relations have perfect recall on their 128 training triples. Only R_{sem} generalizes to the 64 subjects excluded from relation training (0.995 ± 0.009 model recall), and an affine map fitted on the trained subjects identifies held-out targets with accuracy 0.89 ± 0.06 for R_{sem} versus 0.02 ± 0.02 for R_{rnd}. The mechanistic audit in this version explains why. The relation token supplies the final-position *query*, which selects one of two attention readouts; the subject supplies the *value*, which carries the answer content. Attribute training gives all subjects a shared f-geometry, so the semantic readout transfers. Repeated random triples additionally reshape only the relation-trained subject embeddings, so the random readout recalls those subjects but has no information about unseen random assignments. Crucially, a target-stratified affine probe *within the trained subjects* decodes R_{rnd} at 0.987 ± 0.009, and replacing GELU by the identity leaves its recall at 1.000. Thus the earlier claim that R_{rnd} is "stored non-linearly" was too strong: the experiment demonstrates *structure-supported affine transfer versus non-generalizing memorization*, not that the memorized forward computation is intrinsically nonlinear.

2 Data

Vocabulary (265 tokens). A separator `x`; two relation tokens `r_0`, `r_1`; 192 subject tokens `a_0`, ..., `a_191`; 32 object tokens `b_0`, ..., `b_31`; 32 attribute tokens `f_0`, ..., `f_31`; 6 attribute tokens `g_0`, ..., `g_5`.

Entities and attributes. Subject `a_i` carries $(f_{i \bmod 32}, g_{\lfloor i/32 \rfloor})$. Thus the 192 subjects realize each of the 32×6 attribute combinations exactly once. Object `b_j` carries $(f_{(j+3) \bmod 32}, g_0)$.

Sentences. All training sentences have five tokens:

attribute sentence	`x a_i f_p g_q x`	and	`x b_j f_p g_q x`
relation triple	`x a_i r_r b_t x`		

Relations.

$$R_{\text{sem}}(a_i) = b_{f(i)}, \qquad R_{\text{rnd}}(a_i) = b_{\rho(i)},$$

where $f(i) = i \bmod 32$ is the subject's own f-attribute index — so the semantic target is the object whose f-attribute is the subject's, shifted by 3 — and ρ is a seed-dependent balanced random assignment onto the same 32 objects, independent of the attributes. Concrete example (seed 19): `a_37` has attributes (`f_5`, `g_1`), so R_{sem}(`a_37`) = `b_5` (whose attribute sentence reads `x b_5 f_8 g_0 x`), while R_{rnd}(`a_37`) = `b_27`. The corresponding training sentences are `x a_37 r_0 b_5 x` and `x a_37 r_1 b_27 x`.

Split and exact frequency matching. For every `f_`-value, 4 of its 6 subjects are *trained* and 2 are *held out* (128 trained, 64 held-out subjects; seed-dependent). ρ is constructed balanced within both groups. Consequently, under *both* relations: every trained subject occurs in exactly one triple per relation; every object is the target of exactly 4 trained and 2 held-out subjects; both relations contribute exactly 128 training triples, each repeated the same number of times (F). The two relations are therefore indistinguishable by any frequency statistic; only the assignment semantics differ. Held-out subjects appear in attribute sentences but never in any relation triple. These invariants are asserted by `tests/test_linear_shift.py`.

3 Model and training

Model	causal transformer, 1 layer, hidden 64, 4 heads, context 8
Embeddings	input and output embeddings untied; all weights $\mathcal{N}(0, 0.02^2)$
	no attribute-based initialization; nothing frozen; no loss reweighting
Parameters	84,288
Corpus	every attribute sentence $\times A$=128 (= 28,672 sentences)
	+ 128 triples per relation, each $\times F$ ($F \in \{1, 4, 16, 64\}$)
Training	**exactly one epoch**: a single pass over the shuffled corpus
Steps	452 (F=1) to 704 (F=64) steps at batch size 64
Objective	untweighted next-token cross-entropy on all positions
Optimizer	AdamW, constant LR 10^{-3} after 5 linear warmup steps,
	weight decay 0, gradient clip 1.0
Seeds	19, 31, 47 (control data assignment, initialization, and data order)

At F=1 every relation triple occurs exactly once in the entire training history of the model.

4 The metric

A single analysis metric is used, as an operationalization of "the relation is an affine transformation":

> Fit $\hat{W}, \hat{b}$ by ordinary least-squares linear regression from the subject's *input embedding* $s_i \in \mathbb{R}^{64}$ to the target object's *output embedding* (unembedding row) $o_{R(i)} \in \mathbb{R}^{64}$, using the 128 trained pairs. For each of the 64 held-out subjects, predict $\hat{W}s_j + \hat{b}$ and select the nearest object row by cosine similarity among all 32 objects. Report the fraction of held-out subjects for which the selected object is $R(j)$. Chance is $1/32 \approx 0.03$.

This replaces all top-k and accuracy-style analysis metrics of earlier drafts. The identification step avoids the earlier ambiguity objection: for every subject, exactly one object is correct under

each relation. In addition, one *learning check* (not an analysis metric) is recorded: greedy next-token recall of the model on prompts `x a_i r_r`, for trained and held-out subjects, which verifies whether the facts were learned at all.

What the held-out probe can identify. The probe asks whether the target rule fitted on four members of every f-bucket transfers to the other two members. For R_{sem}, this is meaningful: the held-out target is fixed by the same f-attribute visible in the training group. For R_{rnd}, however, the held-out label is sampled independently and never occurs in a relation sentence. No mechanism—linear or nonlinear—can infer that unseen assignment. Chance performance therefore proves absence of transferable random-target information, but by itself does *not* prove nonlinear storage of the 128 observed random facts. Section 6 adds a target-stratified cross-validation restricted to relation-trained subjects specifically to distinguish these claims.

5 Results

		probe accuracy (held-out)		recall (trained)		recall (held-out)	
	F	R_{sem}	R_{rnd}	R_{sem}	R_{rnd}	R_{sem}	R_{rnd}
	1	$1.000 \pm .000$	$.031 \pm .013$	.029	.010	.016	.010
	4	$1.000 \pm .000$	$.016 \pm .013$	.055	.034	.057	.031
	16	$0.990 \pm .007$	$.036 \pm .007$	.672	.167	.635	.031
	64	$0.891 \pm .056$	$.016 \pm .022$	**1.000**	**1.000**	.995	.031
control: no triples		$1.000 \pm .000$	$.021 \pm .015$	–	–	.000	.000

Table 1: Mean ± sd over seeds 19/31/47, at A=128 attribute-sentence occurrences. F is the number of occurrences of each of the 128 triples per relation. Recall columns are the learning check.

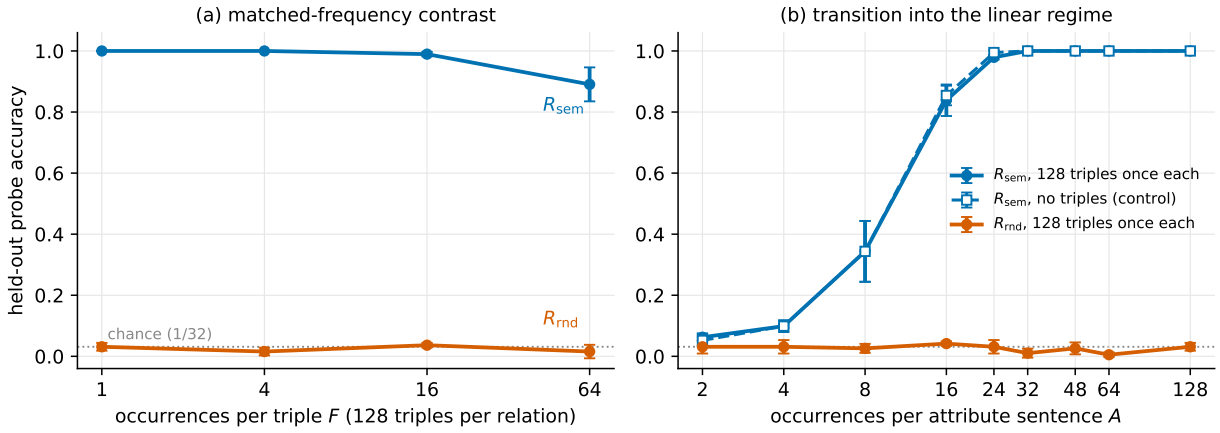


Figure 1: (a) Held-out probe accuracy vs. occurrences per triple: the semantic target transfers through shared f-geometry, whereas independent unseen random assignments do not. (b) Probe accuracy at F=1 as a function of attribute-sentence occurrences A. The identical curve with 128 once-occurring triples (solid) and no triples (dashed) shows that this transition is formation of the attribute scaffold, not learning of the relation forward pass.

Finding 1: with matched frequency, semantics decides whether the learned rule generalizes (Table 1, F=64). Both relations reach perfect recall of all 128 training triples from data with identical frequency statistics. The affine probe transfers to held-out subjects for R_{sem} (0.891 ± 0.056), and the model itself generalizes behaviorally (held-out recall 0.995). Both measures are at chance for the independently assigned held-out targets of R_{rnd}. The supported conclusion is therefore about *transfer*: the attribute-supported rule extends to new members of a known f-class, while random facts are retained only for subjects on which they were trained.

Finding 2: a single-occurrence probe score is evidence for scaffold geometry, not yet for a learned relation. At F=1 the probe is already at 1.000 for R_{sem}, but the model recalls only 0.029 of the corresponding trained facts. The post-hoc regression receives the relation labels and can map the already clustered subject embeddings to object rows even when the transformer cannot answer the prompt. The probe therefore detects that the f-attribute scaffold *supports* an affine semantic rule; it does not establish that one exposure has installed that rule in the forward pass.

Finding 3: the apparent nonlinear→linear transition is the formation of attribute geometry. Sweeping the number of occurrences A of each attribute sentence at F=1 (Fig. 1b) produces a sharp transition of the R_{sem} probe from chance to ceiling ($0.05 \to 0.10 \to 0.34 \to 0.85 \to 1.00$ for $A = 2, 4, 8, 16, 24$). The control with *zero* relation triples produces the same curve. This is decisive evidence that the curve measures subject attribute geometry: the regression itself learns the semantic subject→object association from its labeled fit pairs. At high F, unlike at F=1, behavioral recall shows that the transformer has learned to use this geometry in its own forward pass. Section 6 identifies that computation.

6 Mechanistic account of the two forward passes

6.1 Scope and intervention protocol

The mechanistic analysis targets the only setting in which both tasks have actually been learned: F=64, A=128, k=128. The three reported models are reconstructed by deterministic retraining with seeds 19, 31, and 47; each reconstruction exactly reproduces the corresponding metrics in Table 1. For every prompt, all tensors in the single transformer block are then computed explicitly. An ablation removes one named write at the final prompt position and recomputes layer normalization, the MLP, final layer normalization, and logits. A patch replaces one tensor by the corresponding tensor from a source run and likewise recomputes every downstream operation. This is an interchange intervention on the model's computation, not a probe correlation.

The principal comparisons deliberately use different evaluation groups. For R_{sem} we report the 64 *held-out* subjects, because semantic generalization is the behavior to explain. For R_{rnd} we report the 128 *trained* subjects, because those are the random facts the model knows. Random held-out performance is included as the negative control.

6.2 The exact computation

For a recall prompt $[\texttt{x}, \texttt{a_i}, \texttt{r_r}]$, let $z_p^0 = E[x_p] + P_p \in \mathbb{R}^{64}$ be the token-plus-position residual at position p. The answer is read from the final position $t = 2$. Before attention, z_t^0 contains the relation and position but no subject information. In head $h \in \{0, 1, 2, 3\}$, whose head dimension is

16,

$$q_t^h = W_Q^h \,\mathrm{LN}_1(z_t^0), \qquad k_p^h = W_K^h \,\mathrm{LN}_1(z_p^0), \qquad v_p^h = W_V^h \,\mathrm{LN}_1(z_p^0), \tag{1}$$

$$\alpha_{tp}^h = \frac{\exp(q_t^h \cdot k_p^h/\sqrt{16})}{\sum_{j=0}^{2} \exp(q_t^h \cdot k_j^h/\sqrt{16})}, \qquad A_t = \sum_{h=0}^{3} W_O^h \sum_{p=0}^{2} \alpha_{tp}^h v_p^h, \tag{2}$$

$$z_t^A = z_t^0 + A_t. \tag{3}$$

The 256-unit MLP and readout are

$$u_t = W_1 \,\mathrm{LN}_2(z_t^A) + b_1, \qquad a_t = \mathrm{GELU}(u_t), \tag{4}$$

$$M_t = W_2 a_t + b_2, \qquad z_t^1 = z_t^A + M_t, \tag{5}$$

$$\ell_t = U \,\mathrm{LN}_f(z_t^1), \qquad \hat{x}_{t+1} = \arg\max_v \ell_{t,v}. \tag{6}$$

Here $U \in \mathbb{R}^{265\times 64}$ is the untied unembedding. Every arrow and residual path in Figure 2 corresponds directly to one of these operations.

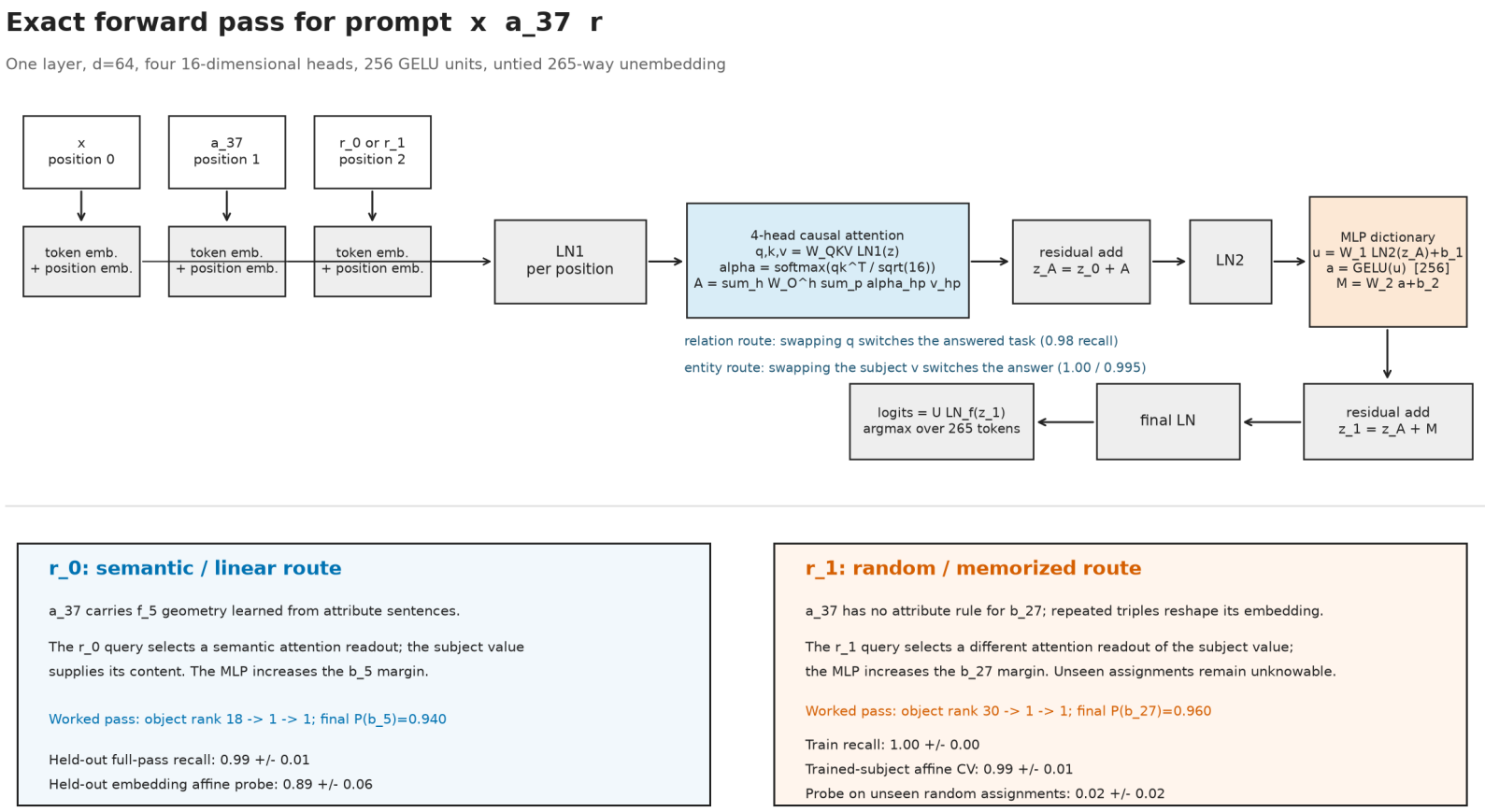


Figure 2: The exact one-block forward computation and the two relation-specific routes. The worked example is seed 19, trained subject a_{37}, for which $R_{\text{sem}}(a_{37}) = b_5$ and $R_{\text{rnd}}(a_{37}) = b_{27}$. The equations show all learned transformations and residual additions. The blue causal annotations summarize the query and value interchange interventions in Table 4.

6.3 Where the answer first appears

Applying the actual final layer norm and unembedding to intermediate residuals gives the logit-lens results in Table 2. "Relation embedding" decodes z_t^0; "after attention" decodes z_t^A; "after MLP" is the model's ordinary output. The relation embedding alone is at chance because it is identical for every subject queried with that relation. Attention creates almost the complete answer: 0.807 semantic recall on unseen subjects and 0.924 random recall on trained subjects. The MLP removes most remaining errors and increases the mean correct-object margin from 1.01 to 3.05 for semantic held-out prompts and from 1.46 to 4.51 for random trained prompts.

decoded stage	R_{sem}, held out	R_{rnd}, trained	R_{rnd}, held out
relation embedding z_t^0	$.031 \pm .000$	$.021 \pm .018$	$.021 \pm .018$
after attention z_t^A	$.807 \pm .104$	$.924 \pm .071$	$.021 \pm .018$
after MLP z_t^1	$.995 \pm .009$	$1.000 \pm .000$	$.031 \pm .016$

Table 2: Full-vocabulary top-1 recall at successive points in the actual forward pass, mean ± sd across three seeds.

For the concrete seed-19 prompt in Figure 2, b_5 has object rank $18 \to 1 \to 1$ and full-vocabulary probability $.014 \to .715 \to .940$ under R_{sem}. Under R_{rnd}, b_{27} has rank $30 \to 1 \to 1$ and probability $.005 \to .435 \to .960$. Thus both answers are already selected by attention; the MLP primarily sharpens them.

6.4 Causal component tests

Table 3 establishes necessity. Removing the entire attention write reduces both relevant behaviors to chance. More specifically, retaining attention but subtracting only the projected value contribution from the subject position also reduces both to chance. In contrast, decoding before the MLP preserves most answers. Replacing GELU by the identity, i.e. using $W_2 u_t + b_2$, preserves essentially all semantic behavior and every trained random fact. GELU is therefore not the essential source of the operational linear/nonlinear contrast in this model. The trained MLP is important for robustness and margin, but its evaluation-time nonlinearity is not necessary for top-1 recall.

intervention	R_{sem}, held out	R_{rnd}, trained
none (ordinary forward pass)	$.995 \pm .009$	$1.000 \pm .000$
zero complete attention write A_t	$.031 \pm .000$	$.031 \pm .000$
zero complete MLP write M_t	$.807 \pm .104$	$.924 \pm .071$
replace GELU by identity	$.990 \pm .018$	$1.000 \pm .000$
remove subject-position value write	$.047 \pm .016$	$.034 \pm .005$

Table 3: Causal interventions with all downstream operations recomputed. Entries are full-vocabulary recall, mean ± sd over seeds.

6.5 The relation is carried by the query; the subject is carried by the value

The cleanest causal factorization comes from patching attention's q, k, and v tensors separately. For a fixed subject, switching only the final-position query q_t from the other relation makes the model answer the other relation with 0.975 recall, averaged over both swap directions. Switching the

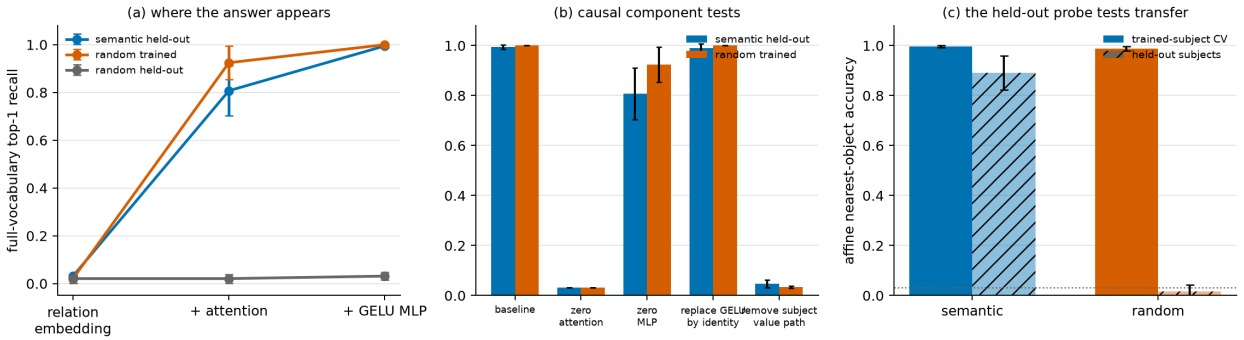


Figure 3: Mechanistic evidence. (a) Attention writes most of the answer and the MLP completes it. (b) Attention and its subject-value path are necessary; GELU itself is not. (c) The original held-out probe tests transfer to subjects with no relation examples. A target-stratified four-fold affine test among relation-trained subjects decodes both mappings, including the random mapping.

relation token's self-key or self-value has no effect. Conversely, for a target-changing permutation of subjects under a fixed relation, switching only the subject value v_1 makes the output follow the source subject; switching only its key k_1 leaves the base answer unchanged. These effects replicate in all seeds (Table 4).

patch	base-target recall	source-target recall
Relation patch, fixed subject; average of $R_{\text{sem}} \leftrightarrow R_{\text{rnd}}$ *directions*		
none	1.000	.029
swap relation query q_t only	.030	.975
swap relation self-key k_t only	1.000	.029
swap relation self-value v_t only	1.000	.029
swap q_t, k_t, v_t	.029	.999
Subject patch, fixed relation; every source has a different target		
swap subject key k_1 only, R_{sem} / R_{rnd}	1.000/.995	.000/.000
swap subject value v_1 only, R_{sem} / R_{rnd}	.000/.000	1.000/.995
swap subject key and value, R_{sem} / R_{rnd}	.000/.000	1.000/1.000

Table 4: Interchange interventions on trained prompts, aggregated over three seeds. "Source-target recall" asks whether the patched run produces the answer belonging to the tensor's source run.

This identifies a compact algorithm:

1. The final-position relation embedding is normalized and projected to a relation-specific query q_t. This query controls which heads attend to the subject and by how much.

2. The subject embedding is normalized and projected to four value vectors v_1^h. These values contain the answer-bearing subject representation. The learned W_O^h projections turn the query-selected mixture into an object-predictive residual write.

3. The MLP transforms and amplifies this write. Its 256 post-GELU activations are strongly target-selective (mean target $\eta^2 = .932$ for R_{sem} and .850 for R_{rnd}), but the information is distributed: linearizing GELU does not destroy it, and no single neuron is the lookup.

4. Final layer normalization rescales the completed residual, and the untied unembedding row with the largest dot product is emitted.

The individual head numbers are not stable across random initializations, but the routing pattern is. For the worked seed-19 subject a_{37}, the attention weight on the subject position in heads 0–3 is $(.389, .036, .959, .816)$ under R_{sem} and $(.927, .740, .166, .029)$ under R_{rnd}. Correspondingly, ablating head 2 hurts semantic held-out recall most among heads 0–2, while ablating head 1 reduces random trained recall from 1.000 to .172. Other seeds permute which head plays each role. Query patching, which does not assume a head correspondence across seeds, recovers the same relation-switching effect in every model.

6.6 Why the two routes generalize differently

Semantic route. Attribute sentences repeatedly train every subject—including the 64 relation-held-out subjects—to predict its f and g attributes. This organizes input embeddings primarily by f: among relation-trained subjects, the mean cosine similarity is $.662 \pm .021$ for pairs with the same semantic target and $.034 \pm .003$ for pairs with different targets. The R_{sem} query selects heads that read this shared geometry from the subject values. Because a held-out a_i with the same f lies in the same geometry even when its g differs, the attention write transfers (.807 recall before the MLP), and the MLP raises it to .995.

Random route. The balanced random label is independent of f and g. Its repeated relation triples nevertheless backpropagate into the embeddings of the 128 trained subjects. Those embeddings acquire a weaker, superposed random-target geometry: same-random-target pairs have mean cosine $.214 \pm .018$, compared with $.045 \pm .004$ for different-random-target pairs. The R_{rnd} query selects a different head mixture that reads this trained-subject signal. Attention consequently recalls .924 of trained random facts before the MLP and all of them after it. A held-out subject received no gradient containing its random target, so its value vector carries no such signal; the same route is at chance. This is ordinary memorization in learned embeddings plus a relation-conditioned readout, rather than evidence for a separate intrinsically nonlinear lookup table.

Direct check of the storage claim. For each target object there are four trained subjects under each relation. Holding out one per target, fitting the same affine embedding→unembedding regression on the remaining three, and rotating through four folds gives 0.997 ± 0.005 accuracy for R_{sem} and 0.987 ± 0.009 for R_{rnd}. Thus both sets of *observed* facts are affinely decodable across trained subjects. The original 0.016 random score arises only when evaluation moves to subjects whose independent random assignments were never presented. Figure 3c visualizes this distinction.

6.7 Mechanistic conclusion

Both tasks use the same circuit template:

$$\boxed{\text{relation token} \to q_t \to \text{head routing}} \quad + \quad \boxed{\text{subject token} \to v_1 \to \text{answer content}}$$
$$\longrightarrow \boxed{\text{MLP margin amplification} \to U}.$$

They differ in the provenance and coverage of the value-space geometry. The semantic signal is learned from attribute sentences for every subject and therefore extrapolates. The random signal is written by relation gradients only into trained-subject embeddings and therefore memorizes. This is the full forward-pass explanation at the component and pathway level supported by causal interventions; it also explains why exactly matching frequencies does not equalize generalization.

7 What this does and does not show

- **Shown behaviorally:** two frequency-matched relations are both memorized on their training triples, but only the attribute-supported rule generalizes to subjects excluded from relation training. Attribute exposure, not triple exposure, produces the geometry detected by the F=1 probe.
- **Shown mechanistically at F=64:** the relation query causally selects the task-specific attention route, the subject value causally supplies answer identity, attention performs most of both readouts, and the MLP amplifies their margins. The semantic route reads a scaffold geometry available for all subjects; the random route reads a weaker target geometry written only into trained-subject embeddings.
- **Not shown:** intrinsically nonlinear storage of the trained random facts. The trained-subject affine cross-validation and GELU-linearization intervention contradict that stronger claim. The held-out random test measures transfer to independent, unobserved assignments and cannot distinguish nonlinear storage from the complete absence of target information.
- **Not shown:** that relation triples cause the F=1 probe transition. The zero-triple control in Figure 1b shows the opposite. At F=1 the model does not reproduce the individual semantic or random facts (recall $\approx .03$), so this setting has no learned recall forward pass to explain.
- **Superseded:** the "single-exposure phase shift" reported in earlier drafts of this document (probe rising $0.27 \to 0.95$ during a single-exposure phase) is an artifact and should not be cited: its regression was fitted on a growing set of pairs (10 at the first snapshot, >500 later) over embeddings that had been *initialized* from the attribute geometry. With a fixed-size probe and standard initialization the effect disappears. All earlier experiments using `structured_embedding_init`, targeted-row updates, pilot-triple repetition, or top-k metrics are superseded by the present setup.

8 Reproduction

Implementation: `experiments/factual_recall/linear_shift.py` (self-contained: world, corpus, training, probe; ∼450 lines). Design invariants: `experiments/tests/test_linear_shift.py`. From `experiments/`:

```
# Table 1 / Figure (a): one seed per command (19, 31, 47), ~1 min each on one GPU
uv run python -m factual_recall.linear_shift --world category --probe embedding \
  --seeds 19 --k-values 0,128 --occurrences 1,4,16,64 --attr-repeats 128 \
  --device cuda --output-dir factual_recall/results/final_category_seed19

# Figure (b): attribute-exposure sweep at F=1, per seed and A
uv run python -m factual_recall.linear_shift --world category --probe embedding \
  --seeds 19 --k-values 0,128 --occurrences 1 --attr-repeats 8 \
  --device cuda --output-dir factual_recall/results/final_category_arsweep_seed19_ar8
# ... repeated for A in {2,4,8,16,24,32,48,64,128}

# Aggregation and figure
# factual_recall/results/final_summary/summary.json, make_figure.py

# Design invariants
```

```
uv run pytest tests/test_linear_shift.py -q

# Mechanistic audit / Figures 2--3: deterministic reconstruction of all three seeds
uv run python -m factual_recall.linear_shift_mechanistic_analysis \
  --seeds 19,31,47 --device cuda \
  --output-dir factual_recall/results/final_mechanistic_analysis \
  --figure-dir paper/figures
```

Raw per-run metrics (including probe R^2/cosine and losses, recorded but not reported) are in `factual_recall/results/final_category_*/linear_shift_metrics.json`. Every cached-stage metric, attention weight, intervention, patch, embedding-geometry statistic, neuron statistic, and trained-subject cross-validation fold used in the mechanistic section is in `factual_recall/results/final_mechanistic_analysis/mechanistic_analysis.json`.